Pre-analysis plan “Many Daughters”

Hypotheses

H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes).*

- That is, denoting X as a measure of support for gender equality,
- (X for respondents with daughters) > (X for respondents without daughters).

H2. *Having daughters leads to more left-leaning political preferences.*

- That is, denoting Y as a measure of left-leaning political preferences,
- (Y for respondents with daughters) > (Y for respondents without daughters).

H3. *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).*

- That is, denoting Z as a measure of gender equality in the household,
- (Z for respondents with daughters) > (Z for respondents without daughters).

H4. *Having daughters leads to a larger increase in the support of gender equality for men*

than for women (with gender equality measured by attitudes).

- That is, denoting X as a measure of support for gender equality,
- (X for male respondents with daughters − X for male respondents without daughters) > (X for female respondents with daughters − X for female respondents without daughters)

Data cleaning

Generally, we adjust the data cleaning procedure to the specific hypotheses and will, therefore, have different workflows for different analyses. We specify which hypothesis we are referring to in each step. If no specific hypothesis is specified, the data cleaning step will be used for all analyses.

We plan to do the following steps for data cleaning:

1. Include only data from years in which the dependent variable is observed (i.e., we do not account for variables that have been observed in any previous year).
2. Case-wise deletion under the missing at random assumption
3. Check for implausible values. Specifically,
   For testing H3: The number of hours in a day should not exceed 24 hours, and the hours per week should not exceed 168 hours. Corresponding observations will be excluded.
4. For testing H3, we include only households for which we have data for both men and women. This means that we only include households with children (according to hgtyp1hh)

Data transformations

We conduct the following data transformations:

- Winsorizing the number of children (variable: *biobirth::biokids*) at the upper 1%
- Use factor analysis to transform the multi-item scale for support for gender equality to a latent variable (see details below)

Data matching

To test H3, we will match the data from individuals and their opposite-sex partners living in the same household (see details below).

Constructs and measures for the main hypotheses

Independent variable: Having daughters

We operationalize having daughters as a dichotomous variable with
- 1 = having at least one daughter
- 0 = having only sons

We infer whether a respondent has a daughter or only sons from the variable gender of the child/children (variable: *biobirth::kidsex01...19* for which female is coded 2)

Rationale: We estimate causal effects by comparing respondents who have at least one daughter to those who have only sons, restricting the sample to individuals with children. This comparison ensures that treatment assignment (child sex) is plausibly random and that we are not comparing parents to non-parents, which would introduce selection bias.

Dependent variables

X: Support for gender equality (as measured by attitudes) [used for H1 and H4]

We use a multi-item measure to capture support for gender equality as an attitude using the position of women scale with the following variables (all from the *pl* dataset):

Variable name + Link	**Description**
plj0621	Stellung der Frau: Arbeiten als Frau
plj0622	Stellung der Frau: Arbeiten als verheiratete Frau
plj0623	Stellung der Frau: Wenn Frau mehr verdient als ihr Partner, Probleme [-> reverse coded]
plj0624	Stellung der Frau: Ausbildung des Sohnes wichtiger als die der Tochter [-> reverse coded]
plj0625	Stellung der Frau: Ehemann sollte das letzte Wort haben [-> reverse coded]

Link to the scale: https://paneldata.org/soep-core/instruments/soep-core-2018-pb-m345-erst/Q466

We use a factor analysis to transform the five manifest variables into one latent variable using the factor function (principal factor) in STATA. We estimate factor scores to get a latent variable for gender equality. Higher values of the factor score indicate a higher support for gender equality.

Y: Left-leaning political preferences

We use the pl/plh0004 to measure left-leaning political preferences (no additional transformations).

Z: Gender equality in the household (as measured by behaviors or outcomes)

We measure gender equality in the household using the ratio of time that men vs. women spend per week on childcare and chores.

Household share of women = (household time spent by women) / (household time spent by women + household time spent by men)

For this function:

- A value of 0.5 indicates an equal share
- A value greater than 0.5 indicates a higher temporal load for women
- A value less than 0.5 indicates a higher temporal load for men

To transform this measure of women's share at household tasks into an estimate for gender equality in the household, we take the absolute difference of the *Household share of women* to 0.5.

Household equality = |*household share of women - 0.5*|

This yields an estimate in which:

- A value of zero 0 indicates that the women/men share is 50% (equality)
- A value that is larger than 0 indicates less equality. (Specifically: if women spent more time, e.g., |0.8-0.5| = 0.3; if women spent less time, e.g., |0.2-0.5| = 0.3)
- A value of 0.5 indicates the highest inequality (either men or women doing all the household tasks)

Multiplying the *Household equality* with -1 yields an estimate in which higher values correspond to more equality (lower values to lesser equality).

We calculate the *household time spent by women* (respectively, *household time spent by men*) as the weekly sum of hours spent on childcare and chores as indicated by the following variables:

Variable name + Link	Description
pli0019_h	Kinderbetreuung Std., Samstag [harmonisiert] (i.e., Childcare, Saturday)
pli0012_h	Hausarbeit Std., Samstag [harmonisiert] (i.e., Chores, Saturday)
pli0022_h	Kinderbetreuung Std., Sonntag [harmonisiert] (i.e., Childcare, Sunday)
pli0016_h	Hausarbeit Std., Sonntag [harmonisiert]

	(i.e., Chores, Sunday)
pli0044_h	Kinderbetreuung Std., Werktg. [harmonisiert] (i.e., Childcare, Weekdays)
pli0043_h	Hausarbeit Std., Werktg. [harmonisiert] (i.e., Chores, Weekdays)

Since this ratio can only be calculated if opposite-sex couples share a household, we match the participants using the following variables:

Variable name + Link	Description
pgpartnr	Unveraend. Personen-id des Partners (i.e., stable personal ID of the partner)
hid	Aktuelle Haushaltsnummer (i.e., current household number)

Control variables

We use the following variables as control variables for the regressions.

Control variables will be included as fixed effects.

Construct	**Variable name + Link**	**Description**	**Transformation or selection for analysis**	**Model**	**Level (dataset)**
Generic, individual control variables					
Gender	pla0009_h	Geschlecht [harmonisiert] (i.e., gender)	Select only male and female	H1, H2, H4	Individuum (pl)
Age	ple0010_h	Geburtsjahr [harmonisiert] (i.e. year of birth)	Transformed to *age* = *survey year* (variable syear) - *year of birth* (ple0010_h)	H1, H2, H4	Individuum (pl)
Marital Status	pgfamstd	Marital Status	Transform to *married* vs. *not married*	H1, H2, H4	Individuum: generated

					personal data (pgen)
Education	pgisced11	ISCED-2011-Klassifikation (i.e., highest education)	none	H1, H2, H4	Individuum: generated personal data (pgen)
Employment	pgemplst	Employment Status	Transform to *full-time* vs. *not full-time* for H1, H2, and H4 For H3: calculating if partners have the same status concerning working *full-time* vs. *not full-time* into categories: *both work equally*, *man works full-time*, *woman works full-time*	H1, H2, H3, H4	Inidviduum: generated personal data (gen)
Generic household-level control variables					
Residence (East vs. West)	hgnuts1	NUTS-Systematik-1 (Bundesland) (i.e., federal state)	Transform hgnuts1 to *east* (Berlin, Brandenburg, Mecklenburg-Vorpommern, Sachsen, Sachsen-Anhalt, Thüringen) vs. *west* (i.e. Baden-Württemberg, Bayern, Bremen, Hamburg, Hessen, Niedersachsen, Nordrheinwestfalen, Rheinland-Pfalz, Saarland, Schleswig-Holstein)	H1, H2, H3, H4	Household: generated household data (hgen)
Household size	PL4	Anzahl aller Haushaltsmitglieder (einschließlich Kinder) (i.e., number of people in the household)	Winsorize at top 1%	H1, H2, H3, H4	Household (hl)
Net income (household)	hlc0005_h	Monatliches HH-Netto-Einkommen	We categorize the household income in	H1, H2,	Household (hl)

		(Euro) [harmonisiert] (i.e., monthly household net income)	steps of 1000€, starting with “less than 1000€,” “1000€ to 1999€,” etc.until “10 000€ or more”	H3, H4	
Living area size	hlf0019_h	Wohnflaeche insgesamt in qm [harmonisiert] (i.e., size of the living area)	Winsorize at top 1%	H3	Household (hl)

Data structure

We match respondents via their personal ID (pid), which is nested in households (hid, current household) and in years (syear). [FYI: Links to pl dataset added as examples].

Regression Specification

Hypothesis 1:

Our Approach:

The outcome variable for Hypothesis 1—support for gender equality—is observed only once per respondent in the SOEP. This precludes the use of panel or Difference-in-Differences methods. We instead leverage the sex of the children as a natural experiment, building on the established assumption that child sex at birth is as good as randomly assigned (e.g., Dahl & Moretti, 2008; Washington, 2008).

This quasi-random variation allows us to estimate causal effects by comparing respondents who have a daughter to those who have only sons. To improve precision and reduce residual variance, we include a set of pre-treatment covariates in a covariate-adjusted linear regression model.

The model we estimate is:

```
reg gender_attitude has_daughter control_vars, vce(cluster hh_id)
```

where

- `gender_attitude` is the outcome variable defined in the constructs section
- `has_daughter` is the treatment variable defined in the constructs section
- `control_vars` are control variables defined in the constructs section

Standard errors are clustered at the household id level because outcomes may be correlated within households.

We restrict the sample as defined in previous sections.

Interpretation of Results: The main estimand is the average treatment effect, interpreted as the effect of having daughters on gender attitudes relative to what would have occurred absent this event. We consider positive and statistically significant at the 5% significance level as support for Hypothesis 1.

Hypothesis 2 & 3:

Our Approach:

We employ the Callaway & Sant'Anna (2021) Difference-in-Differences (DiD) estimator to identify the causal effect of having a daughter on key outcome variables. This method is well-suited for our context for several reasons.

First, the German Socio-Economic Panel (SOEP) provides a longitudinal, unbalanced panel in which individuals are observed over time, but not necessarily in all waves. Childbirth occurs at different times for different respondents, resulting in staggered treatment adoption — i.e., individuals become treated (by having a daughter) in different years. Standard two-way fixed effects (TWFE) DiD estimators are known to be biased under staggered treatment timing when treatment effects are heterogeneous across cohorts or over time (Goodman-Bacon, 2021; de Chaisemartin & d'Haultfoeuille, 2020).

The Callaway & Sant'Anna estimator, implemented in Stata via the `csdid` command, directly addresses this concern by:

- Estimating group-time average treatment effects (ATT_{g,t}), where each treatment group is defined by the year of first exposure (i.e., birth of a daughter)
- Using never-treated individuals as valid comparison groups, avoiding negative weighting biases in traditional TWFE. Never treated is defined as in the Constructs section
- Supporting doubly robust estimation (via inverse probability weighting and outcome regression), which increases robustness to model misspecification

This approach aligns well with our goal: to estimate how individuals' attitudes or behaviors change after becoming the parent of a daughter, relative to an appropriate counterfactual evolution over time.

A. Hypothesis 2:

We define treatment as the first year a respondent becomes the parent of a daughter. The outcome is left leaning *political preferences*, drawn from SOEP survey items as defined in the constructs section.

If respondents have more than one daughter, treatment timing is based on the first daughter's birth.

The causal model is estimated as follows:

```
csdid depvar indepvar, ivar(varname) time(year) gvar(treat_year) ///
        method(dripw) cluster(varname) ///
        wboot reps(#) rseed(42)

estat simple
```

Where:

- `political_preference` is the outcome / depvar
- `indepvar` are the control variables defined in the constructs section
- `year` is the calendar year of the survey
- `id` is the respondent identifier
- `treat_year is` the group-level treatment year, e.g., year when someone first has a daughter
- `method(dripw)` enables doubly robust estimation

Standard errors are clustered based on treat_year allow for dependence within treatment cohorts

We restrict the sample as defined in previous sections.

Interpretation of Results: The main estimand is the average treatment effect on the treated (ATT), interpreted as the effect of having a daughter on political preferences relative to what would have occurred absent this event. We consider positive and statistically significant ATT at the 5% significance level as support for Hypothesis 2.

B. Hypothesis 3:

We define treatment as the first year a respondent in a household becomes the parent of a daughter. The outcome is left leaning *gender equality in the household*, drawn from SOEP survey items as defined in the constructs section.

If household have more than one daughter, treatment timing is based on the first daughter's birth.

The causal model is estimated as follows:

```
csdid depvar indepvar, ivar(varname) time(year) gvar(treat_year) ///
```

```
        method(dripw) cluster(varname) ///

        wboot reps(#) rseed(42)


estat simple
```

Where:

- `gender_equality` is the outcome / depvar
- `indepvar` are the control variables defined in the constructs section
- `year` is the calendar year of the survey
- `id` is the householdidentifier
- `treat_year is` the group-level treatment year, e.g., year when a household first has a daughter
- `method(dripw)` enables doubly robust estimation

Standard errors are clustered based on treat_year allow for dependence within treatment cohorts

We restrict the sample as defined in the previous sections.

Interpretation of Results: The main estimand is the average treatment effect on the treated (ATT), interpreted as the effect of having a daughter on gender equality relative to what would have occurred absent this event. We consider positive and statistically significant ATT at the 5% significance level as support for Hypothesis 3.

Hypothesis 4:

Our Approach:

The outcome variable for Hypothesis 4—support for gender equality—is observed only once per respondent in the SOEP, which rules out panel-based methods. As in Hypothesis 1, we leverage the sex of the children as a natural experiment, under the assumption that child sex is as good as randomly assigned at birth.

To test whether the effect of having a daughter differs by respondent gender, we estimate a covariate-adjusted linear regression model with an interaction term between treatment status and respondent gender.

The model we estimate is:

```
reg gender_attitude i.has_daughter##i.male control_vars, vce(cluster hh_id)
```

where `i.has_daughter##i.male` expands to:

- main effect of has_daughter
- main effect of male
- interaction term has_daughter × male

and

- `gender_attitude` is the outcome variable defined in the constructs section
- `has_daughter` is the treatment variable defined in the constructs section
- `control_vars` are control variables defined in the constructs section

Standard errors are clustered at the household id level because outcomes may be correlated within households.

We restrict the sample as defined in previous sections.

The main estimand for Hypothesis 4 is the interaction term between has_daughter and male (has_daughter × male), which captures whether the effect of having a daughter on gender attitudes differs between men and women. Specifically, it represents the additional effect for male respondents relative to female respondents. A positive and statistically significant coefficient on the interaction term ($p < 0.05$) provides support for Hypothesis 4.

References

Callaway, B., & Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of econometrics*, *225*(2), 200-230.

Dahl, G. B., & Moretti, E. (2008). The demand for sons. *The review of economic studies*, *75*(4), 1085-1120.

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Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of econometrics*, *225*(2), 254-277.

Washington, E. L. (2008). Female socialization: How daughters affect their legislator fathers' voting on women's issues. *American Economic Review*, *98*(1), 311-332.